A MIXED-NORM CAUCHY–SCHWARZ QUESTION OF SAH, SAWHNEY, STONER AND ZHAO

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Abstract. This note records two statements refuted in the *Count ex Machina* archive: each is quoted as posed and answered by an explicit witness, and the archive carries a certificate that recomputes the refutation by rigorous interval arithmetic or exact arithmetic.

Summary: Extension of the mixed-norm Cauchy–Schwarz inequality to all real $1 \le p \le q$ (Sah et al., 2020) *Certificate:* Nonnegative integer $A, B \in \mathbb{Z}_{\ge 0}^{3\times 2}$ at $p = q = 3/2$; the deficit is an integer combination of integer square roots, negative by exact algebraic evaluation.

Summary: Rank-two completely-positive mixed-norm interpolation at $(s, q) = (6/5, 6)$ *Certificate:* Explicit 21-vector Gram matrix and interval-separated ratio > 1.

Posed by A. Sah et al (2020), and put to S. Sra by Y. Zhao in private communication, May 2018; counterexample found by GPT-5.6 Sol (Pro) (2026-07-13); formalized by S. Sra; audited by S. Sra.

Theorem 3: Posed by A. Sah et al (2020), and put to S. Sra by Y. Zhao in private communication, May 2018; counterexample found by Opus 5 (2026-08); formalized by S. Sra; audited by S. Sra.

Theorem 5: Posed by A. Sah et al (2020), and put to S. Sra by Y. Zhao in private communication, May 2018; counterexample found by GPT-5.6 Sol (Pro) (2026-07-13); formalized by S. Sra; audited by S. Sra.

For a matrix $X \in \mathbb{C}^{m\times n}$ the $\ell_{p,q}$ *mixed norm* is defined as:

$$\|X\|_{p,q} = \left(\sum_{j=1}^{n} \left(\sum_{i=1}^{m} |x_{ij}|^p\right)^{q/p}\right)^{1/q}.$$

A matrix is *completely positive* if it has the factorization YY^T for some entrywise nonnegative Y.

Problem 1 (As posed: A. Sah, M. Sawhney, D. Stoner and Y. Zhao, the open question stated immediately after inequality (3.1) in §3)**.** We do not know if the inequality can be extended to $\|A^{\mathsf{T}}B\|_{p,q}^2 \le \|A^{\mathsf{T}}A\|_{q,q}\|B^{\mathsf{T}}B\|_{p,p}$ for all reals $1 \le p \le q$ (this is true for positive integer p by a tensor-power argument).

Problem 2 (As posed, paraphrased: A. Sah, M. Sawhney, D. Stoner and Y. Zhao, the same open question at $B = A$; shared with S. Sra by Y. Zhao in private communication, May 2018)**.** Taking $B = A$ in the question above and writing $Z = AA^{\mathsf{T}}$ for the resulting completely positive matrix: for all reals $1 \le s \le q$ and every completely positive Z,

$$\|Z\|_{s,q}^2 \le \|Z\|_{s,s}\|Z\|_{q,q}.$$

Both Problems 1 and 2 fail by witnesses of quite different character. Problem 1 fails at every non-integer exponent as a consequence of the Hadamard-power theory of FitzGerald and Horn

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From the *Count ex Machina* archive of counterexamples, https://github.com/suvrit/count-ex-machina, whose companion paper is arXiv:2608.29595.

(1977) and Horn and Mathias (1990, Theorem 4.3); that argument is recorded as Remark 4 below (and it is my own observation, not a connection due to an AI assistant). Theorem 3 adds a minimal explicit witness though. The second witness is larger and interval-certified; it refutes the harder $B = A$ specialization of Problem 2. We note that for $p = 1$ (and hence every integer valued p by tensoring) the alleged inequalities do hold—*see Appendix A for details and additional results.*

Theorem 3 ([**Disproof**]: explicit disproof, $A \neq B$)**.** *Let $p = q = 3/2$ and put*

$$A = \begin{bmatrix} 2 & 4 \\ 5 & 0 \\ 1 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 5 \\ 4 & 2 \\ 1 & 0 \end{bmatrix} \in \mathbb{Z}^{3\times 2}_{\geq 0}.$$

Then $\|A^\mathsf{T}B\|^2_{p,q} > \|A^\mathsf{T}A\|_{q,q}\|B^\mathsf{T}B\|_{p,p}$.

Proof. Both matrices are entrywise nonnegative, so they satisfy the hypothesis of the question as posed, not merely the weaker variant in which only the three products are required to be nonnegative. Those products are

$$A^\mathsf{T}A = \begin{bmatrix} 30 & 8 \\ 8 & 16 \end{bmatrix}, \qquad B^\mathsf{T}B = \begin{bmatrix} 17 & 8 \\ 8 & 29 \end{bmatrix}, \qquad A^\mathsf{T}B = \begin{bmatrix} 21 & 20 \\ 0 & 20 \end{bmatrix}.$$

At $p = q = 3/2$ we calculate $\|M\|^{3/2}_{3/2,3/2} = \sum_{ij} |m_{ij}|^{3/2}$, and the deficit

$$\Delta := \|A^\mathsf{T}A\|^{3/2}_{3/2,3/2} + \|B^\mathsf{T}B\|^{3/2}_{3/2,3/2} - 2\|A^\mathsf{T}B\|^{3/2}_{3/2,3/2}$$

is an integer combination of square roots of integers:

$$\Delta = 64 + 64\sqrt{2} + 30\sqrt{30} + 17\sqrt{17} + 29\sqrt{29} - 42\sqrt{21} - 160\sqrt{5}.$$

Deciding the sign of Δ is exact algebra over a real field, with no rounding anywhere:

$$\Delta = -5.150045302282868288532184137690370433502833719528\ldots < 0.$$

Hence $2\|A^\mathsf{T}B\|^{3/2}_{3/2,3/2} > \|A^\mathsf{T}A\|^{3/2}_{3/2,3/2} + \|B^\mathsf{T}B\|^{3/2}_{3/2,3/2}$, and the AM-GM inequality bounds the right side below by $2(\|A^\mathsf{T}A\|_{3/2,3/2}\|B^\mathsf{T}B\|_{3/2,3/2})^{3/4}$. Canceling and raising to the power $2/3$,

$$\frac{\|A^\mathsf{T}B\|_{3/2,3/2}}{\|A^\mathsf{T}A\|^{1/2}_{3/2,3/2}\|B^\mathsf{T}B\|^{1/2}_{3/2,3/2}} = 1.00629360763001504125\ldots > 1. \qquad \square$$

Remark 4 (Why non-integer exponents fail). Let $s \notin \mathbb{N}$ with $s < n - 2$. By the sharpness half of the FitzGerald–Horn theorem (FitzGerald and Horn, 1977) there is a positive semidefinite $Z \in \mathbb{R}^{n\times n}_{\geq 0}$ whose Hadamard power $Z^{\odot s}$ is not positive semidefinite, so $\sigma^\mathsf{T}Z^{\odot s}\sigma < 0$ for some σ. Rescaling if necessary, we can take σ to be in $\{\pm 1\}^n$; then, splitting the index set along the sign of σ write Z as the block Gram matrix of two nonnegative blocks A, B, and the displayed quadratic form is exactly $-\Delta$. This is the mechanism of Horn and Mathias (1990, Theorem 4.3), and it produces a counterexample for *every* non-integer s.

Theorem 5 ([**Disproof**]; computer-assisted disproof $A = B$ case)**.** *There is an explicit 21×21 completely positive matrix Z of rank two with*

$$\|Z\|^2_{s,q} > \|Z\|_{s,s}\|Z\|_{q,q}, \qquad (s, q) = (6/5, 6).$$

Proof. Let X have 21 rows, grouped as follows. For each group use the indicated multiplicity m_g, weight w_g, and direction y_g:

g	1	2	3	4
m_g	1	18	1	1
w_g	46	17	42	1
y_g	$(1, 0)$	$(1, 1/50)$	$(1, 3/50)$	$(0, 1)$.

Every row in group g is $w_g^{5/6} y_g$, and set $Z = XX^{\mathsf{T}}$. Thus $Z \succeq 0$, Z is entrywise nonnegative, and $\operatorname{rank} Z = 2$; being a Gram matrix of nonnegative vectors, Z is completely positive. Put

$$R_Z = \frac{\|Z\|_{6/5,6}}{\sqrt{\|Z\|_{6/5,6/5}\|Z\|_{6,6}}}.$$

Direct interval evaluation gives

$$R_Z \geq 1.00000061733911154365777590600087863798,$$
$$R_Z \leq 1.00000061733911154365777590600087863800.$$

The entire interval lies above one. The source package regenerates Z from the four groups and evaluates the ratio independently at 100 digits; a Sage interval script supplies outward rounding. □

Appendix A. Positive results for Sah et al's mixed-norm CS inequality

This appendix collects the positive results that bound where the mixed-norm counterexamples of Theorems 3 and 5 can live. Throughout we use the convention of Sah et al. (2020),

$$\|X\|_{p,q} := \Big(\sum\nolimits_i \Big(\sum\nolimits_j |x_{ij}|^p\Big)^{q/p}\Big)^{1/q},$$

which is the transpose of the convention used in the reported counterexamples; on the symmetric matrices $A^{\mathsf{T}}A$, $B^{\mathsf{T}}B$ and at $p = q$ the two agree, which covers every statement below. Write p' for the conjugate exponent, $1/p + 1/p' = 1$. The question is whether

$$\|A^{\mathsf{T}}B\|_{p,q}^2 \;\leq\; \|A^{\mathsf{T}}A\|_{q,q}\,\|B^{\mathsf{T}}B\|_{p,p} \tag{A.1}$$

holds for all real $1 \leq p \leq q$ and all entrywise nonnegative A, B.

Lemma A.1. *Let M be Hermitian and $q \geq 1$. Then, $\sup_{\|w\|_{q'} \leq 1} w^* M w \leq \|M\|_{q,q}$.*

Proof. Since $\|w\|_{q'} \leq 1$ iff $\|w\|_{q'}^2 \leq 1$, and $\|w\|_{q'}^2 = \|ww^*\|_{q',q'}$,

$$\sup_{\|w\|_{q'}\leq 1} w^*Mw = \sup_{\|ww^*\|_{q',q'}\leq 1} \operatorname{tr}(Mww^*) \leq \sup_{\|W\|_{q',q'}\leq 1} \operatorname{tr}(MW) = \|M\|_{q,q},$$

the last step being duality of $\|\cdot\|_{q,q}$ and $\|\cdot\|_{q',q'}$. □

Proposition A.2. *Let A, B be entrywise nonnegative and $q \geq 1$. Then (A.1) holds at $p = 1$.*

Proof. As $A^{\mathsf{T}}B \geq 0$ entrywise, $\|A^{\mathsf{T}}Be_i\|_1 = \mathbf{1}^{\mathsf{T}}A^{\mathsf{T}}Be_i$, so by duality

$$\|A^{\mathsf{T}}B\|_{1,q} = \sup_{\|u\|_{q'}\leq 1} \sum_i u_i \mathbf{1}^{\mathsf{T}}A^{\mathsf{T}}Be_i = \sup_{\|u\|_{q'}\leq 1} \langle A\mathbf{1},\, Bu\rangle \leq \sup_{\|u\|_{q'}\leq 1} \|A\mathbf{1}\|_2 \|Bu\|_2,$$

by Cauchy–Schwarz. Now $\|A\mathbf{1}\|_2 = (\mathbf{1}^{\mathsf{T}}A^{\mathsf{T}}A\mathbf{1})^{1/2} = \|A^{\mathsf{T}}A\|_{L_{1,1}}^{1/2}$, while

$$\sup_{\|u\|_{q'}\leq 1} \|Bu\|_2 = \sup_{\|u\|_{q'}\leq 1} (u^{\mathsf{T}}B^{\mathsf{T}}Bu)^{1/2} \leq \|B^{\mathsf{T}}B\|_{q,q}^{1/2}$$

by Lemma A.1. □

This result appears as Lemma 3.1 in Sah et al. (2020); the duality proof recorded there was supplied by S. Sra, and the paper's acknowledgment records as much.

Proposition A.3 (every positive integer p)**.** *If $p \in \mathbb{N}$ and $q \geq p$, then (A.1) holds for all entrywise nonnegative A, B.*

Proof. The block matrix $\left[\begin{smallmatrix} A^\mathsf{T}A & A^\mathsf{T}B \\ B^\mathsf{T}A & B^\mathsf{T}B \end{smallmatrix}\right] = [A\ B]^\mathsf{T}[A\ B]$ is positive semidefinite, so by the Schur product theorem its p-th Hadamard power is too, and is therefore $\left[\begin{smallmatrix} C^\mathsf{T}C & C^\mathsf{T}D \\ D^\mathsf{T}C & D^\mathsf{T}D \end{smallmatrix}\right]$ for some C, D with $C^\mathsf{T}D$, $C^\mathsf{T}C$, $D^\mathsf{T}D$ entrywise nonnegative. Writing $r = q/p \geq 1$,

$$\|A^\mathsf{T}B\|_{p,q} = \Big(\sum_i \Big(\sum_j (A^\mathsf{T}B)_{ij}^p\Big)^{q/p}\Big)^{1/q} = \|C^\mathsf{T}D\|_{1,r}^{1/p} \leq \big(\|C^\mathsf{T}C\|_{1,1}^{1/2}\|D^\mathsf{T}D\|_{r,r}^{1/2}\big)^{1/p},$$

by Proposition A.2, and the right-hand side is $\|A^\mathsf{T}A\|_{p,p}^{1/2}\|B^\mathsf{T}B\|_{q,q}^{1/2}$. □

Proposition A.4 ($q = \infty$)**.** *For $1 \leq p$ and any complex A, B of compatible size, $\|A^*B\|_{p,\infty}^2 \leq \|A^*A\|_{p,p}\|B^*B\|_{\infty,\infty}$. Here entrywise nonnegativity is not needed.*

Proof. By duality and Cauchy–Schwarz,

$$\|A^*B\|_{p,\infty}^2 = \max_i \max_{\|z_i\|_{p'}\leq 1} |z_i^*A^*Be_i|^2 \leq \max_i \max_{\|z_i\|_{p'}\leq 1} (z_i^*A^*Az_i)\,(B^*B)_{ii},$$

and Lemma A.1 bounds the first factor by $\|A^*A\|_{p,p}$ and the second by $\|B^*B\|_{\infty,\infty}$. □

The next two statements concern the specialization $B = A$, i.e. the inequality $\|Z\|_{p,q}^2 \leq \|Z\|_{p,p}\|Z\|_{q,q}$ for completely positive Z.

Proposition A.5 (infinite divisibility)**.** *Let $Z \in \mathbb{R}_{\geq 0}^{n\times n}$ be positive semidefinite and suppose the Hadamard power $Z^{\odot p} = [z_{ij}^p]$ is positive semidefinite for some $p \geq 1$. Then $\|Z\|_{p,q}^2 \leq \|Z\|_{p,p}\|Z\|_{q,q}$ for every $q \geq p$.*

Proof. Write $Z^{\odot p} = X^\mathsf{T}X$. Then $\|Z\|_{p,q} = \|X^\mathsf{T}X\|_{1,q/p}^{1/p}$, and Proposition A.2 bounds this by $\big(\|X^\mathsf{T}X\|_{1,1}\|X^\mathsf{T}X\|_{q/p,q/p}\big)^{1/2p} = \|Z\|_{p,p}^{1/2}\|Z\|_{q,q}^{1/2}$. □

Corollary A.6 (order at most three)**.** *Let $Z \in \mathbb{R}^{n\times n}$ be positive semidefinite with $n \leq 3$, with no assumption that its entries are nonnegative. Then $\|Z\|_{p,q}^2 \leq \|Z\|_{p,p}\|Z\|_{q,q}$ for all $1 \leq p \leq q$.*

Proof. For $n \leq 3$ a PSD Z has $|Z| := [|z_{ij}|]$ PSD as well.[1] Since $p \geq 1 \geq n-2$, the FitzGerald–Horn theorem (FitzGerald and Horn, 1977) makes $|Z|^{\odot p}$ positive semidefinite, so Proposition A.5 applies to $|Z|$. The mixed norms depend only on $|z_{ij}|$, so the bound transfers to Z. □

Together, Propositions A.2, A.3, A.4 and Corollary A.6 confine any counterexample to non-integer p, finite q, and order at least four. Theorem 3 lives at $p = q = 3/2$ and order four (as a Gram matrix of two 3×2 blocks), and Theorem 5 at $(6/5, 6)$ and order 21 — in each case the smallest region the positive results leave open.

Code and certificate

The case is the directory `counterexamples/rank-two-mixed-norm/` of the archive, at commit `73b416e`. Its `verify.py` recomputes every refutation in this note using exact arithmetic or rigorous interval enclosures, never floating-point evidence alone; it raises an error if any check fails and otherwise writes the certificate below. Independent Sage interval cross-checks: `verify_mixed_norm.sage`. To rerun it:

```
git clone https://github.com/suvrit/count-ex-machina
cd count-ex-machina/counterexamples/rank-two-mixed-norm
python verify.py
```

- Theorem 3 (result `mixed-norm-general-s`): exact arithmetic.

[1] This is a classical fact about small Hermitian matrices; the source manuscript for this case attributes it to Marcus and Watkins (1969), a reference the present text has not independently checked.

- Theorem 5 (result `yufei-psd`): rigorous interval arithmetic.
- `artifacts/certificate.json`: Construction data of the 21-vector Gram matrix and the high-precision ratio; the Sage script provides the interval certificate.

AI USAGE

The credit line under the title records, for each statement, who posed it, who found its counterexample and when, and who formalized and audited it; the counterexamples here were found by GPT-5.6 Sol (Pro) (2026-07-13); Opus 5 (2026-08). This note was assembled mechanically from the case's source in the archive by `tools/notes.py`, written with Claude Code; its mathematics is exactly that of the archive at the commit cited above.

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